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Using this function, estimate the average salary in 1995. Answer: \$1,943.36 in thousands of dollars, or \$1,943,360.

NT FOR Suppose that three points (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) are given. Upon substituting these into Eq. (6.1) to solve for a , b , and c , what conclusion would you reach if there is no solution to the system of three equations?

Section 6.1 Follow-Up Exercises

In Exercises 1–16, determine which functions are quadratic and identify values for parameters a , b , and c .

1 $f(x) = 4x^2 - 20$

2 $f(x) = \frac{x^2}{4} + 3x$

3 $g(h) = 3h - 450$

4 $h(u) = 24 - u + u^2/3$

5 $g(s) = (s - 4)^2$

6 $f(x) = (3 - 2x + x^2)^2$

7 $f(x) = (x^2 - 4x + 5)/3$

8 $v(s) = s^2 + (s + 5)/3$

9 $h(x) = (x^2 - 2x)^2$

10 $f(x) = 25 - x^3$

11 $f(x) = \sqrt{x^4 - 4x^2 + 4}$

12 $g(h) = 1,000 - h^2$

13 $h(v) = 4v - 5 + 20v^2$

14 $f(x) = x^2 - 2x + x^3$

15 $g(h) = \sqrt[3]{(h - 4)^6}$

16 $f(x) = \sqrt[4]{(2 - x)^8}$

In Exercises 17–32, determine the concavity of the parabola representing the quadratic function, its y intercept, its x intercepts if any exist, and the coordinates of the vertex. Sketch the parabola.

17 $f(x) = -x^2$

18 $f(x) = -x^2 + 6x - 9$

19 $f(x) = x^2 - 3x + 2$

20 $f(x) = x^2 - 4$

21 $f(x) = -x^2 + 5$

22 $f(x) = x^2 - 4x + 4$

23 $f(x) = \frac{x^2}{2} - 10x$

24 $f(x) = \frac{-x^2}{2} + 24x$

25 $f(x) = -x^2 - 9$

26 $f(x) = 3x^2 + 7x - 20$

27 $f(x) = 2x^2 + 4$

28 $f(x) = -3 - x^2$

29 $f(x) = 6x^2 + x - 12$

30 $f(x) = 4x^2 + 5x - 6$

31 $f(x) = -x^2 - 5x$

32 $f(x) = x^2 - 7x + 10$

33 Sketch the following quadratic functions, making note of the values of $|a|$ for each and the relative steepness of the three parabolas.

$$f(x) = x^2 \quad f(x) = 0.01x^2 \quad f(x) = 100x^2$$

34 Determine the equation of the quadratic function which passes through the points $(0, 10)$, $(1, 6)$, and $(-2, 24)$.

35 Determine the equation of the quadratic function which passes through the points $(1, -1)$, $(-3, -33)$, and $(2, -8)$.

CHAPTER 6 QUADRATIC AND POLYNOMIAL FUNCTIONS

